TONER-1

260,032 parameters
int8 weights, one scale per row
Inference runs inside the PostScript interpreter

State of the art among language models that execute inside a print job.

SAMPLE OUTPUT

Unedited model output. No prompt: generation starts at the BOS token.

Once upon a time, there was a lonely drop of ink. The drop was getting shiny and had a soccer ball. He wanted to read the map to the printer. But when he went to a clock, he saw a yellow ball and gave it to his friend. The ball looked at it and wanted to visit him.

The drop of ink showed Rosa the ball and crossed the wrong wing. The printer took it and found the machine inside. The printer was so surprised! It was a…

TOKEN STREAM

All 200 sampled tokens, one box per vocabulary piece. · marks a space, ¶ a newline.

·Once	·upon	·a	·time	,	·there	·was	·a	·l	on	e	l	y	·d	r	o	p	·of	·in	k	.
·The	·d	r	o	p	·was	·g	et	t	ing	·s	h	in	y	·and	·had	·a	·so	c	c	er
·b	all	.	·He	·want	ed	·to	·	re	ad	·the	·m	a	p	·to	·the	·p	r	in	t	er
.	·B	ut	·w	he	n	·he	·w	ent	·to	·a	·c	l	o	ck	,	·he	·saw	·a	·y	e
ll	ow	·b	all	·and	·g	a	ve	·it	·to	·his	·friend	.	·The	·b	all	·l	oo	ked	·a	t
·it	·and	·want	ed	·to	·	v	is	it	·h	im	.	·	¶	T	he	·d	r	o	p	·of
·in	k	·s	h	ow	ed	·	R	o	s	a	·the	·b	all	·and	·c	r	o	s	s	ed
·the	·w	r	on	g	·w	ing	.	·The	·p	r	in	t	er	·to	o	k	·it	·and	·f	ou
nd	·the	·m	a	ch	in	e	·in	s	id	e	.	·The	·p	r	in	t	er	·was	·so	·s
u	r	p	r	is	ed	!	·I	t	·was	·a										

HOW THIS PAGE WAS MADE

1 HOST
Sends the print job. Performs no arithmetic.

→

2 INTERPRETER
Parses the int8 weights, then runs the forward pass once per token.

→

3 PAGE
The same job typesets the story and calls showpage.

TOKENS	SECONDS	TOKENS / SEC	SEED	TEMPERATURE
200	2.4	83.0	40	0.8

INTERPRETER	GPL Ghostscript · revision 10071 · PostScript version 3010
MODEL	stories260K printer-tuned (Llama-2 architecture) · dim 64 · 5 layers · 8 query / 4 KV heads · 512-token vocab
STOPPED BY	Token budget: STEPS = 200. The story was cut off mid-thought.

Host compute: 0 FLOPs. Inference cost includes toner.

Every word above was generated on-device, one token at a time, by the interpreter named in this footer; the host computer only sent bytes. Seconds are interpreter usertime spent generating. rand is implementation-defined, so the same seed on a different interpreter tells a different story.